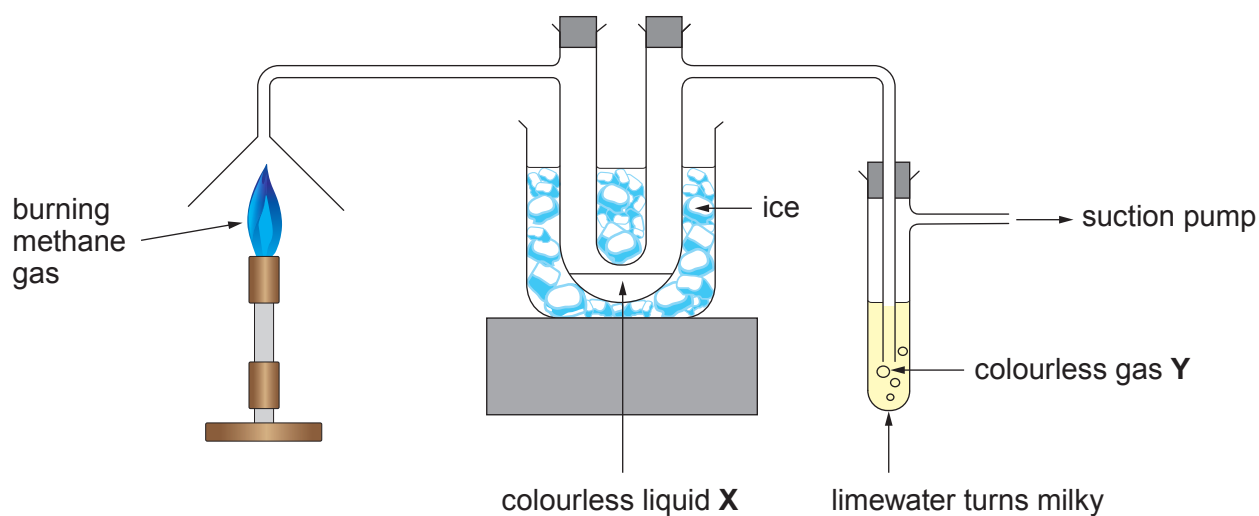


3. (a) The diagram shows apparatus that can be used to investigate the products formed when methane gas burns. Methane has the formula CH_4 .



- (i) Name the gas present in the air that is needed for methane to burn. [1]

.....

- (ii) Name the products formed. [2]

Colourless liquid X

Colourless gas Y

- (iii) Tick (✓) the box that gives the name of the group of substances that methane belongs to. [1]

alkenes ☐

monomers ☐

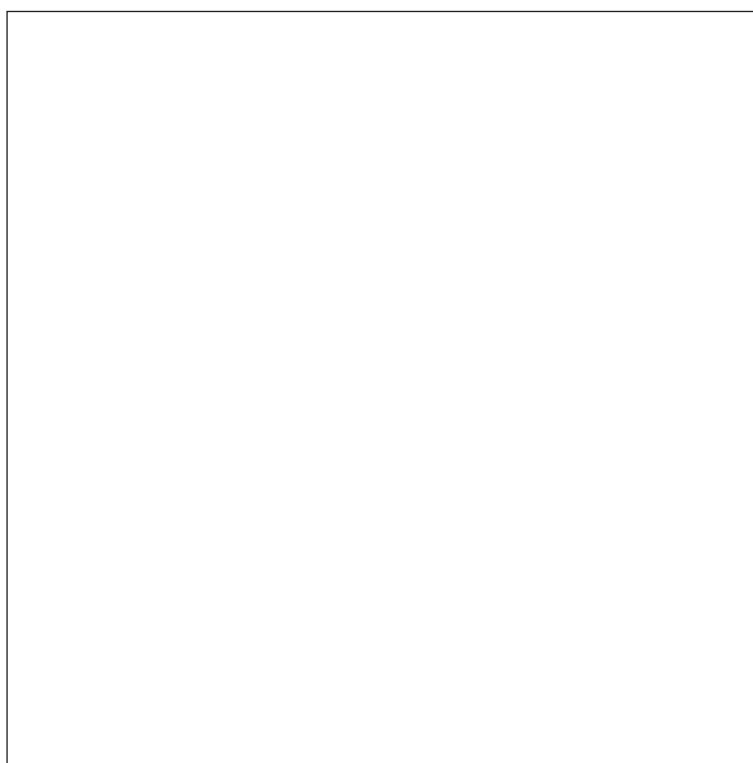
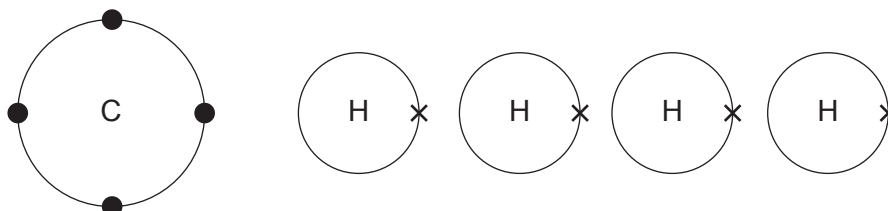
polymers ☐

alkanes ☐



- (b) Methane has the formula CH_4 . Its molecules consist of one carbon atom and four hydrogen atoms bonded together.

- (i) Draw a diagram in the box to show how the atoms bond to form a methane molecule. [2]



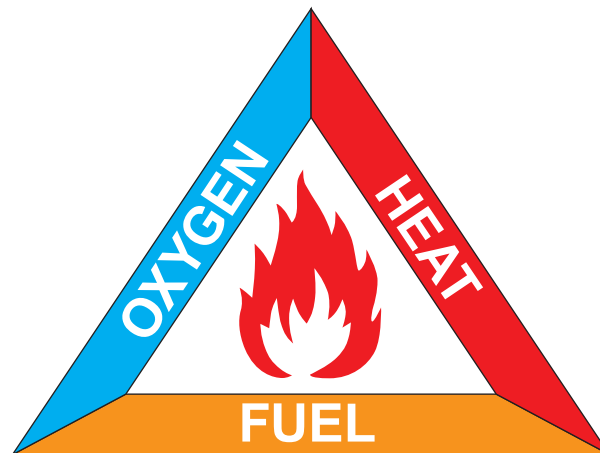
- (ii) Give the name of this type of bonding.

[1]

.....



- (c) The fire triangle can be used to explain how fires are extinguished.



The following table gives information about three different types of fire and the methods used to extinguish them.

However, each of the three fires has an **error** in **one** of the columns.

Fire	Type of fire	Firefighting method	How method works
1	chip pan fire	tea towel	removes the heat
2	bonfire	fire blanket	removes the heat
3	electrical fire	fire breaks	removes the fuel

Circle the errors for fires **1** and **2** and then correct them in the table below.

The error for fire **3** has already been circled and the correction given.

[2]

Fire	Correction
1	
2	
3	Fire breaks are used to extinguish a forest fire

